MILESTONE 1

BIN 381 – Group P4

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TABLE OF CONTENTS

1. Business Understanding
    - 1.1 Project Background & Socioeconomic Decision Problem
    - 1.2 Stakeholders & Information Needs
    - 1.3 Business Objectives & Success Criteria
    - 1.4 Analytics Goals & Technical Success Criteria
    - 1.5 CRISP-DM Methodology
    - 1.6 Inventory of Available Resources, Skills, and Data Assets
    - 1.7 Detailed Risk Register
    - 1.8 Ethical Considerations & Responsible AI Usage Log
    - 1.9 Project Plan: Milestones 2-4
2. Data Understanding
    - 2.1 Full Dataset Portfolio Inspection
    - 2.2 Dataset-Selection Matrix
    - 2.3 Data Description & Key Variables
    - 2.4 Data Relationships & Grain Analysis
    - 2.5 Initial Hypotheses

1. BUSINESS UNDERSTANDING

1.1 Project Background & Socioeconomic Decision Problem

1.1.1 Project Background

South Africa is still struggling a lot with inequality, poverty, and unemployment — this isn't new, but it's still a huge issue. The country's Gini coefficient is one of the highest in the world (around 0.63), and even though there've been a bunch of policy interventions over the years, poverty (looked at from multiple angles, not just income) is still a big problem in both cities and rural areas. The SDHSA project ('Socioeconomic Development & Household Well-being in South Africa') is basically trying to move past just looking at one indicator at a time, and instead figure out how all these different types of household hardship connect to and affect each other.

To do this, we're using data from the General Household Survey (GHS) — a survey Stats SA runs every year that covers things like education, health, housing, access to services, and economic activity. The goal is to use this data to come up with insights that can actually help inform policy decisions.

1.1.2 Socioeconomic Decision Problem

The primary socioeconomic decision problem is:

How can we identify vulnerable communities in South Africa where multiple deprivations overlap (such as poor housing, limited access to services, low income, and food insecurity), in order to come up with solutions that work across multiple sectors?

This problem is significant for the following reasons:

Challenge	Impact
Single-sector interventions often fail when households face multiple challenges	Current programs have limited effectiveness
Resource allocation is currently based on single indicators (e.g., income poverty only)	Inefficient use of the limited public resources
Understanding the *combination* of deprivations enables targeted, efficient policy design	Better outcomes with the same resources if not less

1.1.3 Research Context

This project is based on the idea that poverty isn't just about income - it's multi-dimensional, meaning it also includes sectors like health, education, housing, and access to services. This lines up with South Africa's National Development Plan 2030, which pushes for a more joined-up approach to tackling poverty, so this analysis fits right into that bigger picture.

1.2 Stakeholders & Information Needs

Stakeholder	Role	Information Needs	Decisions Supported
Department of Planning, Monitoring and Evaluation (DPME)	Primary	Identify districts with multiple deprivations for resource prioritization	Budget allocation, cross-departmental coordination
Provincial Government Departments	Secondary	Understand the causes of deprivation in specific provinces	Provincial-level intervention design
Municipalities	Secondary	Localized vulnerability profiles	Service delivery planning, informal settlement upgrading
International Development Organizations (e.g., World Bank, UNICEF)	Tertiary	Evidence for program targeting	Funding allocation, program evaluation
Civil Society Organizations	Tertiary	Community vulnerability patterns	Advocacy, program design

1.3 Business Objectives & Success Criteria

1.3.1 Business Objective 1

The idea here is to create a kind of index or grouping system that shows how households are doing when you look at well-being from multiple angles - not just income, but other factors too.

Business Success Criteria:

We want our typology to spot at least 3 clear vulnerability clusters, and these should match up with the real poverty hotspots we already know about in SA. Basically, when we compare our clusters to Stats SA's municipal-level poverty data, we're aiming for at least 80% agreement - so our results should broadly line up with what's already known. We also need to pull out at least 5 insights that are genuinely useful (and not obvious) for policymakers - like noticing that peri-urban informal settlements often have water but not proper sanitation, compared to rural areas. Lastly, everything needs to be explained simply enough that someone without a technical background could read it in a policy brief and get it.

1.3.2 Business Objective 2

To figure out the key drivers and relationships between indicators that best predict household vulnerability.

Business Success Criteria:
The model needs to identify at least 3 main drivers of food insecurity, plus 2 secondary ones. The findings should be presented in a way that's easy for stakeholders to follow (like a decision tree), showing how demographic characteristics lead to different outcomes. The recommendations need to be specific enough to actually guide budget decisions for at least two provincial departments. And the whole analysis needs to be reproducible and transparent so it can be monitored going forward

1.4 Analytics Goals & Technical Success Criteria

1.4.1 Analytics Goal 1: Clustering

To run an unsupervised clustering analysis that groups households into distinct segments based on their well-being characteristics.

Technical Success Criteria:

Criterion	Target	Measurement Method
Cluster separation	Silhouette score > 0.3	Silhouette analysis
Cluster stability	At least 2 validation methods agree (elbow, gap statistic, silhouette)	Multiple validation techniques

Criterion	Target	Measurement Method
Cluster interpretability	Each cluster must have a clear, meaningful profile	Profiling with descriptive statistics
Statistical significance	Clusters must differ significantly ($p < 0.05$) on key indicators	ANOVA/Kruskal-Wallis tests

1.4.2 Analytics Goal 2: Predictive Modeling

To use statistical methods to figure out the strongest predictors of household food insecurity and economic participation.

Technical Success Criteria:

Criterion	Target	Measurement Method
Model performance	ROC-AUC > 0.75 for binary outcomes OR $R^2 > 0.60$ for continuous	Cross-validation
Variable significance	At least 3 variables with $p < 0.05$	Statistical testing
Model stability	Variable importance stable across at least 2 modeling approaches	Compare methods (e.g., logistic regression vs. random forest)
Interpretability	Model must be explainable to non-technical stakeholders	Use simple models where possible; SHAP values if using complex models

1.5 CRISP-DM Methodology

The project will follow the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology. This structured approach helps ensure.:

Phase	Key Activities	SDHSA Application	Status
1. Business Understanding	Define objectives, assess situation, establish success criteria	Completed in this Milestone	✅
2. Data Understanding	Collect initial data, describe, explore, verify quality	Completed in this Milestone	Done
3. Data Preparation	Clean, construct, integrate, format data	Milestone 2	Pending
4. Modeling	Select modeling techniques, build, assess models	Milestone 3	Pending
5. Evaluation	Evaluate results, review process, determine next steps	Milestone 4	Pending
6. Deployment	Plan deployment, produce final report, create dashboard	Milestone 4	Pending

Why CRISP-DM is appropriate:

. It's an industry-standard methodology for data mining projects

. The iterative approach means we can go back and refine things as our understanding improves

. It puts business understanding first, before jumping into analysis

. It gives us a clear documentation framework so the work stays reproducible

1.6 Inventory of Available Resources, Skills, and Data Assets

1.6.1 Technology & Tools

Category	Specific Tools	Purpose
Data Processing	RStudio, tidyverse, dplyr, janitor	Data manipulation, cleaning, transformation
Data Quality	skimr, naniar, DataExplorer, VIM	Missing data analysis, quality assessment
Statistical Analysis	ggplot2, corrplot, GGally, psych	EDA, correlation, visualization
Business Intelligence	Power BI Desktop	Exploratory dashboard creation
Reproducibility	R Markdown/Quarto	Documented, reproducible analysis notebook
Collaboration	GitHub	Version control, team collaboration

1.6.2 Team Skills

Skill	Team Member(s)	Proficiency Level
Data Wrangling	Person 1, Person 2	Advanced
Statistical Analysis	Person 2, Person 3	Intermediate-Advanced
Machine Learning	Person 3, Person 4	Intermediate
Data Visualization	Person 1, Person 4	Advanced
Power BI	Person 4	Intermediate
CRISP-DM Knowledge	All	Foundational
Ethical Data Handling	All	Foundational

Skill	Team Member(s)	Proficiency Level
Report Writing	Person 1, Person 2	Advanced

1.6.3 Data Assets

Asset	Type	Source	Access	Size (Est.)
D01_Population_Profile.csv	Primary Data	Stats SA GHS	Direct download	~8 MB
D04_Dwelling_Conditions.csv	Primary Data	Stats SA GHS	Direct download	~5 MB
D05_Basic_Services.csv	Primary Data	Stats SA GHS	Direct download	~6 MB
D07_Household_Wellbeing.csv	Primary Data	Stats SA GHS	Direct download	~4 MB
D09_Household_Resources.csv	Primary Data	Stats SA GHS	Direct download	~9 MB
Survey Metadata	Documentation	Stats SA ISIbalo Portal [citation:11]	Direct download	~2 MB

1.7 Detailed Risk Register

ID	Risk Description	Category	Likelihood	Impact	Mitigation Strategy	Contingency Plan
R1	Data Integration Failure: Inability to merge the 5 selected datasets due to missing or inconsistent primary keys (HHID)	Technical	High	High	- Inspect key structures in each dataset during early data understanding - Document any merge issues - Explore alternative merge strategies (e.g., geographic combinations)	Reduce to 2-3 datasets; pivot to single-dataset analysis
R2	Data Quality Issues: High proportion of missing data, especially for income and sensitive variables	Data	Medium	High	- Systematic missing data analysis using naniar and VIM - Assess pattern of missingness (MCAR, MAR, MNAR) - Apply multiple imputation using mice	Remove problematic variables; use complete-case analysis with caution

ID	Risk Description	Category	Likelihood	Impact	Mitigation Strategy	Contingency Plan
R3	Project Scope Creep: Attempting to use too many datasets or address too many research questions	Project Management	Medium	Medium	- Strictly adhere to selected 5 datasets - Define focused research questions at outset - Regular check against project goals	Prioritize core objectives; defer secondary analyses
R4	Ethical Privacy Breach: Accidental disclosure of individual household-level data	Ethics	Low	High	- Store data on secure, university-managed drives - Report at aggregated level only - Check for personal identifiers	Report to ethics committee; implement additional anonymization
R5	Skills Gap: Team members lack specific R skills for advanced analysis	Resource	Low	Medium	- Identify skill gaps early - Schedule paired programming sessions - Leverage	Focus on well-documented packages; simplify analysis plan

ID	Risk Description	Category	Likelihood	Impact	Mitigation Strategy	Contingency Plan
					online resources	
R6	Interpretation Bias: Misinterpreting correlation as causation or ignoring confounding variables	Analytical	Medium	Medium	- Peer-review findings within team - Focus on associations, not causality - Document assumptions clearly	Consult with domain experts; validate with literature
R7	Data Access Restrictions: Special permissions required for some datasets	Operational	Low	Medium	- Download all datasets immediately upon project start - Contact Stats SA user services if inaccessible	Use available datasets; adjust scope
R8	Model Overfitting: Models perform well on training data but fail to generalize	Analytical	Medium	High	- Use cross-validation - Keep models simple and interpretable - Validate on hold-out data	Simplify models; use regularization techniques

ID	Risk Description	Category	Likelihood	Impact	Mitigation Strategy	Contingency Plan
R9	Resource Constraints: Time or computing limitations	Resource	Medium	Medium	- Realistic timeline planning - Use efficient R code	Focus on core analyses; use cloud computing if needed

1.8 Ethical Considerations & Responsible AI Usage Log

1.8.1 Ethical Principles

Principle	Application to SDHSA Project
Privacy	We're only working with anonymized data for the analysis, and we're keeping our reporting at a broader level (like province or municipality) so there's no chance of accidentally identifying specific individual
Bias	We remain aware of potential biases in the data and modeling approaches, and we interpret results with caution when generalizing to marginalized populations.
Transparency	All methodology, assumptions, and limitations fully documented
Reproducibility	All code and analysis reproducible
Beneficence	Project aims to improve policy outcomes for vulnerable populations
Non-maleficence	Ensure findings are not misused to discriminate against communities

1.8.2 Responsible AI Usage Log

Date	Activity	Potential Bias/Privacy Risk	Mitigation Applied	Status
2026-09-07	Data loading and inspection	Risk of identifying individuals through unique household IDs	Confirmed no personal identifiers (names, addresses) in datasets; IDs are anonymized	Done
2026-09-07	Dataset selection	Risk of excluding relevant variables that may disadvantage certain groups	Justified exclusions based on relevance to well-being; all core indicators included	Done
2026-09-07	Initial data quality assessment	Missing data may disproportionately affect certain population groups	Documented missing data patterns by variable; will analyze missingness by demographic group in M2	Done
[TBD]	Model development	Models may perpetuate historical biases	Will test model performance across demographic subgroups; use fairness metrics	Pending
[TBD]	Reporting	Results could be misinterpreted or misused	Will include clear limitations section; present findings in	Pending

Date	Activity	Potential Bias/Privacy Risk	Mitigation Applied	Status
			context; engage stakeholders	

1.9 Project Plan: Milestones 2-4

Milestone	Phase	Key Activities	Deliverables	Timeline	Dependencies
M2	Data Preparation	1. Merge 5 selected datasets on HHID 2. Validate merge (counts, duplicates) 3. Handle missing data (imputation/dropping) 4. Create composite well-being indicators 5. Feature engineering (derived variables) 6. Final data quality assessment	- Cleaned master dataset - Imputation report - New derived variables - Data quality final assessment - Reproducible R notebook	Week 3-4	R1, R2
M3	Modeling	1. Perform cluster analysis (K-means, hierarchical) 2. Determine optimal clusters	- Cluster assignments - Model performance metrics	Week 5-6	M2 completion

Milestone	Phase	Key Activities	Deliverables	Timeline	Dependencies
		(elbow, silhouette, gap) 3. Build logistic regression for food insecurity 4. Build regression for income prediction 5. Model validation (cross-validation) 6. Variable importance analysis	- Variable importance analysis - Model selection justification - Reproducible R notebook		
M4	Evaluation & Deployment	1. Interpret and profile clusters 2. Translate findings to policy recommendations 3. Build final Power BI dashboard 4. Create stakeholder presentation 5. Finalize project report 6. Document limitations and next steps	- Final dashboard - Policy brief - Project documentation - Stakeholder presentation - Final report	Week 7-8	M3 completion

Key Milestone Dependencies:

M2 depends on successful data merge (R1) and manageable missing data (R2)

M3 depends on complete, clean dataset from M2

M4 depends on valid models from M3

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2. DATA UNDERSTANDING

2.1 Full Dataset Portfolio Inspection

The available datasets come from the General Household Survey (GHS), which Statistics South Africa conducts every year. The GHS captures the living circumstances of South African households across several dimensions, including education, health, housing, access to services, and economic activity.[citation:11].

2.1.1 Full Portfolio Summary

Dataset Code	Dataset Name	Indicative Analytical Domain	Initial Assessment	Selection Status
D01	Population_Profile.csv	Population and demographic profile	High relevance - Provides segmentation context	Selected
D02	Learning_Profile.csv	Education and learning profile	Medium relevance - Structural factor	Excluded
D03	Economic_Participation.csv	Economic participation	Medium relevance - Overlaps with D11	Excluded
D04	Dwelling_Conditions.csv	Dwelling and	High relevance - Core well-	Selected

Dataset Code	Dataset Name	Indicative Analytical Domain	Initial Assessment	Selection Status
		housing conditions	being indicator	
D05	Basic_Services.csv	Water, sanitation and basic services	High relevance - Core well-being indicator	Selected
D06	Household_Energy.csv	Household energy	Medium relevance - Overlaps with D05	Excluded
D07	Household_Wellbeing.csv	Food security and household well-being	High relevance - Outcome measure	Selected
D08	Connectivity_Access.csv	Connectivity and digital access	Low relevance - Emerging indicator	Excluded
D09	Household_Resources.csv	Income, grants, assets and resources	High relevance - Economic capacity measure	Selected
D10	Mobility_Access.csv	Mobility and access	Low relevance -	Excluded

Dataset Code	Dataset Name	Indicative Analytical Domain	Initial Assessment	Selection Status
			Specific, not universal	
D11	Labour_Market.csv	Labour-market modelling data	Medium relevance - Too detailed for our current scope	Excluded

2.2 Dataset-Selection Matrix

Dataset	Relevance to Well-being	Data Quality (Expected)	Linkability	Time Period	Selected?	Justification
D01_Population_Profile	High (Context)	High	High	2023	yes	Provides demographic context for segmentation; essential for disaggregated analysis
D02_Learning_Profile	Medium	High	High	2023	no	Structural factor affecting long-term well-being; not a measure

Dataset	Relevance to Well-being	Data Quality (Expected)	Linkability	Time Period	Selected?	Justification
						of *current* household well-being; can be explored indirectly through income/employment analysis
D03_Economic_Participation	Medium	Medium	High	2023	no	Redundant with D11; focus on outcomes (well-being) rather than labor market *processes*
D04_Dwelling_Conditions	High (Core)	High	High	2023	yes	Direct measure of housing deprivation; a key component of multi-dimensional poverty

Dataset	Relevance to Well-being	Data Quality (Expected)	Linkability	Time Period	Selected?	Justification
D05_Basic_Services	High (Core)	High	High	2023	yes	Direct measure of service access; critical for health and quality of life
D06_Household_Energy	Medium	High	High	2023	no	Overlaps with D05 Basic Services; energy access captured in D05
D07_Household_Wellbeing	High (Outcome)	Medium	High	2023	yes	Direct measure of food security; primary outcome variable of interest
D08_Connectivity_Access	Low	Medium	High	2023	no	Emerging indicator; not yet a foundational well-being dimension

Dataset	Relevance to Well-being	Data Quality (Expected)	Linkability	Time Period	Selected?	Justification
D09_Household_Resources	High (Enabler)	Low (Missing Income)	High	2023	yes	Measures economic capacity; essential for understanding resource constraints
D10_Mobility_Access	Low	Medium	High	2023	no	Specific to mobility; not a universal well-being factor
D11_Labour_Market	Medium	Medium	High	2023	no	Too detailed for current scope; overlaps with D03 and D09

2.2.1 Selection Rationale

The 5 selected datasets (D01, D04, D05, D07, D09) represent a balanced portfolio of well-being indicators:

Dataset	Role in Analysis	Justification
D01	Context	Who is affected? Provides demographic break down
D04	Core Indicator	Fundamental need - Housing quality
D05	Core Indicator	Fundamental need - Access to basic services

Dataset	Role in Analysis	Justification
D07	Outcome	What is the result? - Food security and well-being
D09	Enabler	What resources are available? - Income and assets

This selection fits with what we're going for -looking at household well-being from multiple angles - without piling on redundant variables. It keeps the analysis manageable while still covering the main dimensions of deprivation.

2.3 Data Description & Key Variables

2.3.1 Unit of Observation

We're focusing on households as our main unit of analysis. The GHS collects data at both the household and individual level, but for this project, we're sticking to the household level since that's what gives us the clearest picture of overall well-being.

2.3.2 Key Variables of Interest

Dataset	Key Variables	Data Type	Purpose in Analysis
D01	Province, Household Size, Household Head Age, Sex of Household Head, Population Group	Categorical, Numeric	Demographic segmentation; control variables
D04	Dwelling Type (formal/informal/traditional), Tenure Status (owned/rented/rent-free), Roof/Wall Material	Categorical	Housing deprivation indicator
D05	Main Water Source (piped/community/other), Sanitation Facility Type (flush/pit/open), Refuse Disposal Type	Categorical	Service access indicators

Dataset	Key Variables	Data Type	Purpose in Analysis
D07	Food Security Status (moderate/severe), Food Consumption Score, Source of Food	Categorical, Numeric	Well-being outcome measure
D09	Main Income Source, Monthly Income (or income bracket), Ownership of Assets, Receipt of Grants	Numeric, Categorical	Economic capacity measures

2.4 Data Relationships & Grain Analysis

2.4.1 Data Relationships

Dataset	Key	Relationship to Master Dataset	Relationship Type
D01_Population_Profile	HouseholdID	1:1 (one record per household)	Primary
D04_Dwelling_Conditions	HouseholdID	1:1 (one record per household)	Direct
D05_Basic_Services	HouseholdID	1:1 (one record per household)	Direct
D07_Household_Wellbeing	HouseholdID	1:1 (one record per household)	Direct
D09_Household_Resources	HouseholdID	1:1 (one record per household)	Direct

2.4.2 Merge Strategy

All datasets will be merged by a LEFT JOIN on HHID to create a master analytic dataset. This strategy is based on the following assumptions:

HHID is the unique identifier for each household.

There are no duplicate HHID values within a single dataset.

The merge will be validated by checking that the number of rows in the master dataset equals the number of unique HHIDs in the primary dataset (D01).

2.4.3 Merge Validation Steps

Step	Action	Expected Outcome
1	Check HouseholdID exists in all datasets	True
2	Check HouseholdID uniqueness in each dataset	No duplicates
3	Perform LEFT JOIN from D01 to other datasets	Row count = nrow(D01)
4	Validate merge by checking HouseholdID count	All D01 HouseholdIDs present
5	Document any merge failures	Issues resolved or documented

2.4.4 Limitations

If HHID isn't present in all the datasets, we may need to look at alternative keys instead (e.g., geographic identifiers combined with household number). This is a critical data quality check that will be done in R.
Different variable naming conventions across the datasets may also require careful mapping.
We'll also need to verify that time periods align across the dataset.

2.5 Initial Hypotheses

Based on the problem understanding and available data, the following hypotheses will be tested in later milestones:

ID	Hypothesis	Test Method	Variables	Expected Finding
H1	Households in informal dwellings have significantly	Chi-square test of independence	Dwelling Type (D04) vs. Water Source	Informal dwellings will

ID	Hypothesis	Test Method	Variables	Expected Finding
	lower access to piped water and flush toilets compared to formal dwellings		and Sanitation (D05)	show lower service access
H2	Receipt of social grants is positively correlated with food insecurity, indicating grants are insufficient to meet basic needs	Logistic regression	Grant Receipt (D09) vs. Food Security Status (D07)	Grant recipients will have higher food insecurity
H3	Larger household size is associated with lower per-capita income, leading to increased reliance on social grants and subsequent deprivations	Multiple regression + path analysis	Household Size (D01), Income (D09), Grant Receipt (D09), Food Security (D07)	Larger households will have lower per-capita income and more deprivation
H4	Geographic location (province) interacts with dwelling type to predict access to basic services	Two-way ANOVA	Province (D01), Dwelling Type (D04), Service Access (D05)	Rural provinces will show different patterns than urban provinces
H5	Households with formal employment income are more likely to have asset	Mediation analysis	Income (D09), Asset Ownership	Asset ownership will partially mediate

ID	Hypothesis	Test Method	Variables	Expected Finding
	ownership, which mediates the relationship between income and well-being		(D09), Food Security (D07)	income-wellbeing relationship

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